CE205 Database and Information Retrieval

Class 2

Objective: The main objective is to remove the redundant data from the base relation using normalisation process and convert the relation to desired normal form.

Problem 1:

Consider the relation REFRIG(MODEL#, YEAR, PRICE, MANUF_PLANT, COLOR), which is abbreviated as REFRIG(M, Y, P, N, C), and the following set of functional dependencies:

M→N
MY→P
N→C

Compute closures of M, MY, and N. Then, consider the decomposition of REFRIG into sub-relations R1(M, Y, P) and R2(M, N, C). Is this decomposition lossless? Justify your answer using the conditions of lossless decomposition.

Solution guideline for problem 1:

Find the following:

M^+=

$\{MY\}^+$=

N^+=

Justify, lossless or not? *(for conditions see slide 12 of Lec 2b)*

Condition 1:

Condition 2:

Condition 3:

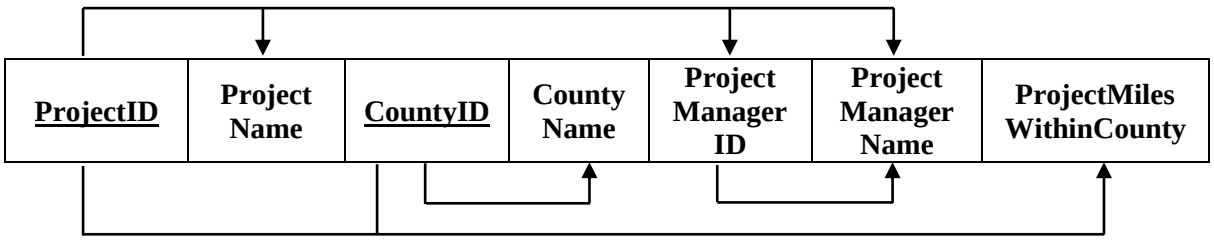
Problem: 2(a)

Consider the relation HIGHWAY with sample data (*the underlined attributes represent the primary key, and assume that there is no space in the attributes' name*).

ProjectID	**Project Name**	**CountyID**	**County Name**	**Project Manager ID**	**Project Manager Name**	**ProjectMiles WithinCounty**
1	Road X	1	Wilson	M1	Bob	10.00
1	Road X	2	Ottawa	M1	Bob	17.00
1	Road X	3	Davis	M1	Bob	12.00
2	Road Y	3	Davis	M2	Sue	23.00
3	Bridge A	1	Wilson	M3	Lee	0.50
3	Bridge A	2	Ottawa	M3	Lee	0.30
4	Tunnel Q	2	Ottawa	M1	Bob	2.00
5	Road W	4	Pony	M4	Rob	23.00

- *The HIGHWAY Project Table captures the data about projects and their length (in miles).*
- *Each project has a unique Project ID and Project Name.*
- *Each county has a unique County ID and County Name.*
- *Each project manager has a unique Project Manager ID and Project Manager Name.*
- *Each project has one project manager.*
- *A project manager can manage several projects.*
- *A project can span across several counties.*
- *This table records the length of the project in a county in the ProjectMilesWithinCounty column*

Based on the above assumptions, the following functional dependencies exist in the relation



(a) Depict full key functional dependencies, partial functional dependencies (if any), and transitive functional dependencies (if any) in the elation HIGHWAY? Provide reasons why these functional dependencies are full key, partial, and transitive?

(b) Redesign the relation HIGHWAY such that it must satisfy 2NF and 3NF? First, show the process of normalizing the relation in 2NF. Then, show the process of normalizing the relation in 3NF. Clearly, show all functional dependencies and keys in the resulting sub-relations.

Solution guideline for problem 2(a):

i. Write the **primary key** of the relation:

ii. Write **prime attributes**:

iii. Write **non-prime attributes**:

iv. Write **proper subsets of the primary key**:

v. Have you identified the **partial FD(s)** in the relation? Why are they partial FD(s)? Give reason.

vi. Have you identified the **transitive FD(s)** in the relation? Why are they transitive FD(s)? Give reason.

vii. Have you identified the **full key FD(s)** in the relation? Why are they full key FD(s)? Give reason.

viii. Decompose the base relation to remove partial FDs **(2NF):**

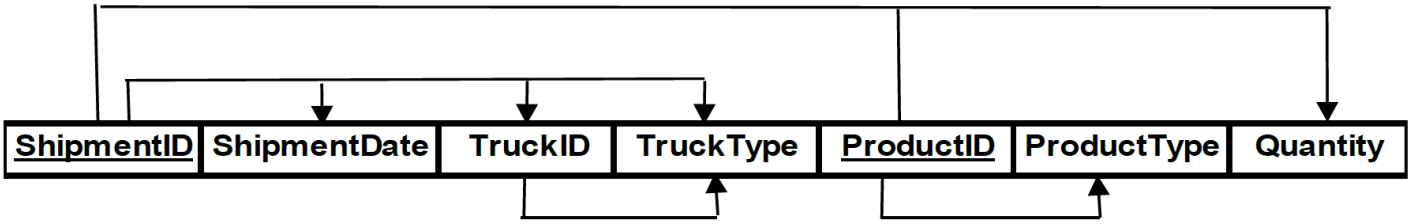
ix. Write the **primary and foreign key(s)** in the decomposed sub-relations(s):

x. Decompose the sub-relation(s) to remove transitive FDs **(3NF):**

xi. Write the **primary and foreign key(s)** of the decomposed sub-relations(s):

Problem 2(b):

Repeat the same steps as mentioned in problem 2(a) for the relation given below:



Solution: *(follow the guideline of problem 2a)*